Chapter 22

Digital Dentistry in Operative Dentistry

"Perfection is achieved, not when there is nothing more to add, but when there is nothing left to take away".

—Antoine de Saint-Exupéry

Digital technology, computerized dentistry and digital dentistry are general terms used to describe the clinical application of computer-assisted design/computer-assisted machining (CAD/CAM). The restorative dentistry application of CAD/CAM technology is the fabrication and delivery of permanent restorations for teeth and implants. For the past 30 years, the incorporation of dental CAD/CAM into direct patient care has provided a way for dentists to deliver esthetic ceramic restorations in a single dental appointment.

There are three sequences involved in the CAD/CAM process.[1]

- An intra-oral scanner or camera is used to accurately record the hard and soft tissue geometry of the patient's intra-oral condition to a computer program in the first sequence. This is commonly referred to as a *digital impression*. Digital impression systems are designed to accurately record the intra-oral geometry and then transmit the files to a dental laboratory for design and fabrication of the desired restoration. There is minimal if any opportunity for the dentist to design any aspect of the restoration on digital impression systems. They were developed to leverage the digital recording process to take advantage of:
  - Comfort and efficiency of not using conventional impression materials.
  - Convenience and accuracy of digital transmission of the case to the dental laboratory.
- A proprietary software design program is used to create a *virtual restoration* (the volume proposal) in the second sequence. The software programs have the capability of controlling and editing the various parameters of the restoration such as emergence profile, proximal contact and occlusal relationships.
- Once the proposal of the restoration has been completed, a computer-controlled device is used to produce the restoration in the third sequence. The most common device uses a subtractive process to machine (i.e. *grind or mill*, depending on whether carbide burs or diamonds are used) the final restoration from a preformed block of a variety of restorative materials.

Understanding these three sequences provides a basis for deciding how best to implement the technology in the dental office and also creates a simple way to categorize various systems in the marketplace.

Chairside CAD/CAM systems employ all three sequences of the CAD/CAM process in the dental office. They also record intra-oral scans but provide in addition a software program for designing restorations as well as milling units to fabricate the restoration during a single dental appointment. They are designed to leverage the efficiency of a single appointment procedure for the delivery of ceramic restorations. Chairside CAD/CAM systems are the primary focus of the chapter because they allow the dentist complete control of the design, fabrication and delivery of the restoration in the office during a single appointment.

Clinical Application

Treatment planning considerations for CAD/CAM restorations are not significantly different from ceramic restorations done with conventional impression materials and techniques. The CAD/CAM system represents an *alternative means* of restoration fabrication, not the restoration per se.

The type of restoration (inlay, onlay, crown), the choice of material to be used, the desired occlusal relationships and ability to isolate the tooth preparation for delivery of the restoration are several primary factors to consider rather than the restoration fabrication process itself. A case in point is that the predictable ability to isolate a subgingival margin for adhesive cementation is a much more

important factor to consider than whether the ceramic restoration is fabricated with a conventional or digital impression technique. Nonetheless, there are a few specific considerations relative to the use of digital impressions.

- The relative size of the camera may be a concern for patients with a restricted ability to open wide. Generally, if there is sufficient vertical space to complete the tooth preparation with a dental handpiece, there is sufficient space for use of a digital camera.
- However, patients with a severe gag reflex may appreciate the use of a digital impression more than conventional impression since there is no physical contact with the intra-oral tissues by a tray or impression material when recording a digital impression.

Chairside CAD/CAM Systems

- Dr Francois Duret conceptualized the first chairside CAD/CAM system in 1973.
- However, the first functioning chairside CAD/CAM prototype was introduced in the 1980s through the collaboration between a Swiss prosthodontist, Dr Werner Mörmann, and an Italian electrical engineer, Marco Brandestini. Dr Mörmann's vision was to use CAD/CAM technology to deliver esthetic ceramic restorations with improved longevity in a single appointment that avoided the deleterious consequences polymerization shrinkage caused in composite restorations.[2,3]
- The CEREC 1 unit marked the introduction of the CEREC system in 1985 with the first clinical trials reported in 1987.[4] The system has evolved through a series of hardware and software innovations and upgrades that culminated with the introduction of the first colour-streaming powder-free intra-oral camera in 2012, the CEREC Omnicam (Fig. 22.1).[5,6]
- *CEREC Omnicam*: The Omnicam is the imaging camera of the CEREC AC acquisition unit. The Omnicam is connected to a chairside computer with the three-dimensional design software and a liquid crystal display (LCD) monitor. The CEREC system is electronically connected to a milling unit. The newest version is a dry grinding and wet milling unit that is the first in-office means by which to dry mill zirconia. Single-visit, full-contour, milled zirconia restorations are then further processed by sintering and glazing the restoration in the CEREC SpeedFire sintering furnace. This is the first in-office zirconia sintering furnace that can sinter preshaded CEREC zirconia material (CEREC Zirconia) in



Figure 22.1 CEREC OmniCam and MCX milling chamber.
(Courtesy: Dentsply Sirona.)



Figure 22.2 PlanFit chairside CAD/CAM system.
(Courtesy: D4D Technologies.)

10–15 min, as compared to the multiple hours that are generally necessary for sintering laboratory processed zirconia restorations.

- *E4D Dentist System*: The *E4D Dentist System* (D4D Technologies) was introduced in 2008 with its DentaLogic software offering a true three-dimensional virtual model.[7] The E4D Dentist System has an intra-oral laser scanner, mobile Design Center with DentaLogic software and a separate milling unit with a dedicated CAM server computer. The milling unit has two opposing electric motors that automatically change between three different diamonds depending on the specifics of the restoration dimensions. The milling unit has a dedicated computer server, which allows independent operation separate from the Design Center (after completion and transmission of the case design) (Fig. 22.2).
- CS 3500: The CS 3500 is a powder-free intra-oral scanner that was introduced in 2013. The Carestream system allows for in-office design using CS Solutions Restore software and in-office fabrication using the CS 3000 milling unit. The digital files may also be uploaded to a dental laboratory via CS Connect as they are compatible with a number of commercial laboratory design programs such as 3Shape (3Shape) and Exocad (Exocad Gmbh). Some of the unique features of the intra-oral scanner include a disposable tip and a guiding light, which indicates a successful scan. The Carestream system has chairside capability for only a limited number of applications, including inlays, onlays and crowns.

Chairside CAD/CAM systems are able to produce inlays, onlays, veneers and crowns. Some systems also have the ability to fabricate short-span fixed partial dentures as well as temporary restorations in the dental office. Additional applications, such as implant abutments, fixed partial dentures and orthodontic appliances, are unique to specific systems (Table 22.1).

Tooth Preparation Principles for CAD/CAM Restorations

Guidelines for tooth preparation for all-ceramic restorations are generally based on the specific geometries and thickness dimensions required to provide optimum strength for the selected ceramic material. For example, the occlusal reduction for a full contour zirconia crown is less

Table 22.1 Clinical applications for chairside CAD/CAM systems

System	Inlays/onlays	Crowns	Veneers	FPDs	Implant abutment	Orthodontic application	CBCT compatible
CEREC (Dentsply Sirona)	X	X	X	X	X	X	X
BlueCam	X	X	X	X	X	X	X
OmniCam							
PlanFit (Planmeca)	X	X	X	X	–	–	X
CS Solutions (Carestream)	X	X	X	–	–	–	X

FPD, fixed partial dentures.

than that required for a leucite-reinforced or feldspathic porcelain crown, even if both materials are fabricated with a chairside CAD/CAM system. However, there are several preparation guidelines that enhance the accuracy of the CAD/CAM restoration due to their influence on the imaging and fabrication process, such as smooth contours, rounded transitions and uniform pulpal floor.

A concept unique to CAD/CAM restorations has been termed '*undermilling*' or '*overmilling*'. Any preparation geometry smaller than the dimension of the milling instruments presents a potential problem. Either the instrument must cut away more restorative material to ensure the restoration seats completely (overmilling) or it does not remove the smaller geometry (undermilling), leaving restorative material that will prevent complete seating of the restoration.

Utilization of a preparation geometry with *smooth contours, gently flowing curves* and *rounded transition angles* helps to limit the potential for overmilling and ensure an accurate internal adaptation of the restoration.

Crowns

Tooth preparation for CAD/CAM-fabricated crowns is essentially the same as for laboratory-fabricated ceramic crowns.

- The margin design for an all-ceramic crown requires a bulk of ceramic at the margin to avoid the risk of ceramic chipping or fracture. This is best accomplished with a *shoulder, sloped shoulder* or *heavy chamfer margin* (Fig. 22.3).
- All of these margin geometries are equally useful for chairside CAD/CAM crowns as long as the *internal angles are rounded* (i.e. not sharp or angular) to facilitate internal adaptation of the crown.
- The occlusal and axial reduction requirements are determined based on the specific material properties of the ceramic material selected for the crown.
- *Axial reduction* for CAD/CAM ceramic crowns should be *1–1.2 mm* for the monolithic ceramic materials as there is no opaque inner coping that needs to be covered (veneered).
- A greater variation in occlusal reduction may be expected as a function of the material selected.
  - Most glass ceramic materials require a minimum of 1.5 mm occlusal reduction in the central fissure area and over nonfunctional cusps and 2 mm over functional cusps (Fig. 22.4).
    - Laboratory-based (in vitro) research studies suggest that full contour zirconia crowns may be fabricated with an occlusal thickness of 1 mm without compromising the strength of the crown.
    - Additionally, the manufacturer of lithium disilicate blocks has recently announced that e.maxCAD crowns now only require a 1-mm occlusal reduction; however, this claim still requires independent validation.
- All angles of the preparation should be rounded to facilitate accurate internal adaptation and needless overcutting of the internal surface of the crowns during milling.
- It is not uncommon to identify significant undercuts in a crown preparation as a result of prior large restorations or caries excavation. *Undercuts* will be accurately recorded by the digital camera and be present in the calculated virtual model in the CAD software program. However, the milling process will not duplicate any internal undercuts in the preparation that might prevent seating of the restoration. *These undercuts would subsequently be filled in with the resin cement.*
- Alternatively, undercuts could be *blocked out* using the software program during design of the restoration. This process is *not recommended*; however, as it could ultimately result in an internal inaccuracy as the digital model is altered. Therefore, if blockout is necessary it is best accomplished with an appropriate dentin substitute material, as indicated, prior to digital scanning.

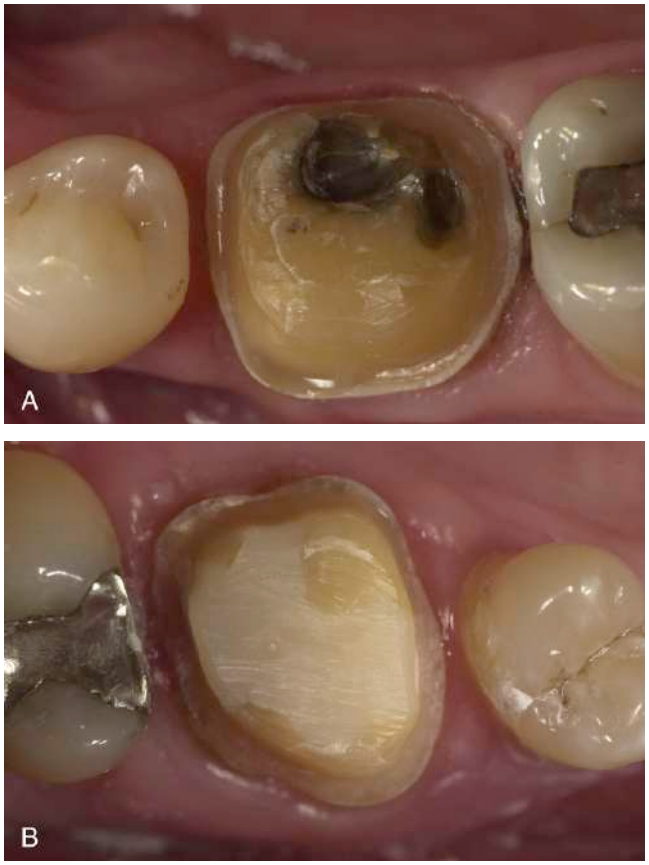


Figure 22.3 (A and B) Examples of chairside CAD/CAM crown preparations.

Inlays and Onlays

CAD/CAM inlay and onlay preparations are primarily adhesive-style preparations that rely on the adhesion of the resin cement to dentin and enamel for retention of the restoration (Fig. 22.5A).



Figure 22.4 (A and B) Examples of the occlusal clearance for chairside CAD/CAM crown preparations.

- These preparations are *divergent* and relatively nonmechanically retentive in design as this provides a more conservative preparation than the requirement for mechanical resistance through grooves, slots or boxes (Fig. 22.6A–C).
- The internal aspect of the preparation should avoid sharp divots or concavities and all *internal angles should be rounded* (Fig. 22.6D).
- *Occlusal reduction* should be uniform and of sufficient thickness (Fig. 22.5B) to provide optimum strength of the selected ceramic material similar to crown preparations. Preparation should allow for:
  - Minimum of 1.5 mm of ceramic thickness in the central fossa and over nonfunctional cusps.
  - Minimum of 2 mm over functional cusps.
- All *cavosurface margins* should be strategically placed away from the contact position of the opposing cusp(s) and be well defined (smooth) for easy identification in the design software.
- *Bevelled margins must be avoided*, as thin areas of ceramic are prone to fracture.[8,9]
- The preparation isthmus should be at least *2 mm in faciolingual width* so as to avoid inlay/onlay fracture (Fig. 22.7).



Figure 22.5 (A) Onlay preparations for teeth #3 and #4 relying primarily on the adhesive bond of the resin cement for retention of the restoration. (B) Typical diamond rotary instruments used for ceramic inlay or onlay tooth preparations.

- CAD/CAM onlay preparations *do not utilize the creation of a ferrule* with the ceramic restoration as used with metal castings (Fig. 22.8). On the contrary, a ferrule may actually interfere with proper seating of the restoration due to undermilling of the relatively parallel walls and/or sharp transitions. Alternatively the system may significantly overmill the intaglio surface of the restoration, thinning the restoration further. Both undermilling and overmilling are detrimental to the final onlay restoration.
- A *butt-joint margin is preferred* as it allows for proper thickness of the ceramic material at the margin, reducing the incidence of marginal fracture (Fig. 22.9). Unfortunately, this marginal design often causes a visible demarcation between the tooth and the restoration. Hence, when the blend of the restoration and the tooth is of esthetic concern, a modification of the facial butt-joint margin is necessary. The cavosurface margin may be modified with a *football-shaped diamond at a 45-degree angle*, creating opportunity for a transition of ceramic thickness over the underlying enamel while maintaining a bulk of ceramic at the margin for physical strength of the ceramic (Fig. 22.10). An in vitro study on ceramic preparations for conservative restoration of endodontically

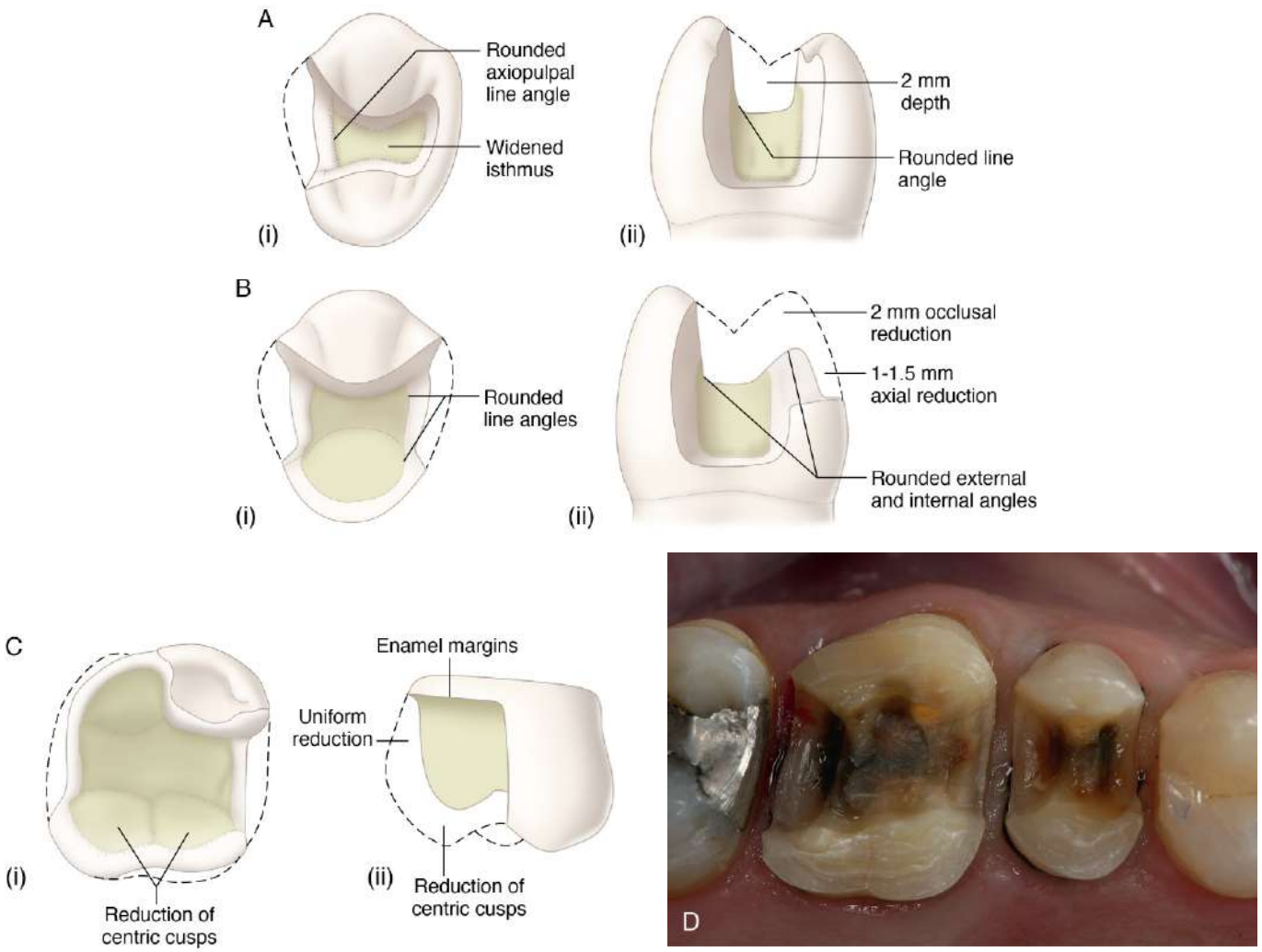


Figure 22.6 (A) (i) Mesio-occlusal (MO) onlay preparation for tooth-coloured inlay in maxillary first premolar (*occlusal view*). Isthmus should be at least 2 mm wide to prevent inlay fracture. The axiopulpal line angle should be rounded to avoid seating errors and to lower stress concentrations. (ii) Mesio-occluso-distal (MOD) onlay preparation for tooth-coloured inlay in the maxillary first premolar (*proximal view*). The pulpal floor should be prepared to a depth of 2 mm, and the axiopulpal line angles should be rounded. The proximal margins should be extended to allow at least 0.5 mm clearance of contact with the neighbouring tooth. Gingival margins in enamel are greatly preferred. (B) Occlusal (i) and proximal (ii) views of mesio-occluso-disto-lingual (MODL) onlay preparation on maxillary first premolar. The lingual cusp has been reduced and the lingual margin extended beyond any possible contact with opposing tooth by preparing a 'collar'. Functional cusps require 2 mm of occlusal reduction. All internal and external line angles are rounded. (C) Mesio-occluso-disto-facio-lingual (MODFL) onlay preparation on the maxillary right first molar. Distofacial (*DF*), mesiolingual (*ML*) and distolingual (*DL*) cusps are reduced. (i) Occlusal view. (ii) Facial view. (D) Full cuspal onlay preparation for a maxillary first molar and MODL onlay for a maxillary second premolar with smooth and rounded internal angles.

treated teeth reported a significant improvement in the marginal and internal fit of the ceramic restoration with this modified-margin design relative to both a ferrule and 90-degree butt-joint margin.[10]

Chairside CAD/CAM Clinical Workflow

The clinical workflow to design and fabricate a chairside CAD/CAM restoration is relatively similar for all currently available systems, with noticeable differences in the unique cameras, software programs and milling chambers of marketed systems (Fig. 22.11A–L workflow sequence).

- The tooth preparation, including the margins, must be visible by retraction of soft tissues and isolated from moisture contamination so that the camera may accurately capture a digital image of the preparation. The opposing arch is also recorded, as well as a scan of the dentition in maximum intercuspation from the facial aspect.
- The computer software virtually articulates the opposing models using the scan of the facial surfaces so the appropriate occlusal relationships for the restoration may be designed.
- The restoration design is initiated by identifying the margins of the planned restoration to identify the limits of the restoration design.

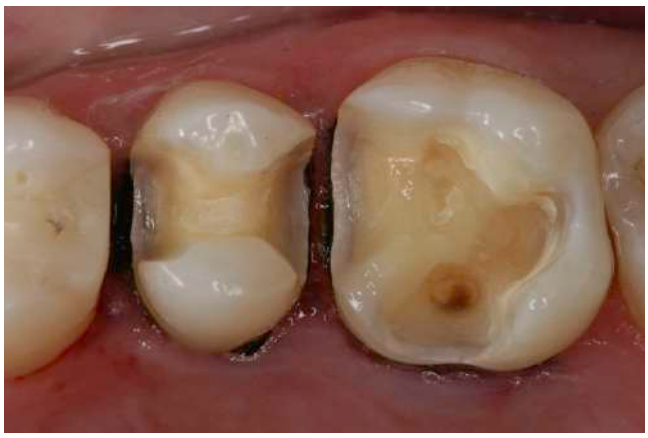


Figure 22.7 Inlay preparation for a maxillary second premolar and onlay preparation for a maxillary first molar with smooth flowing outline form with crisp cavosurface margins.

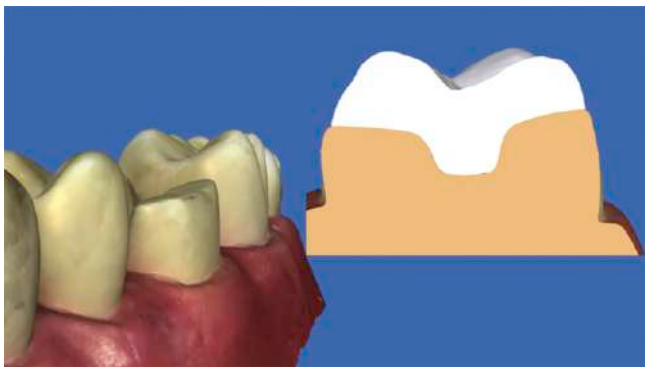


Figure 22.9 Preferred butt-joint margin design for ceramic onlay preparation.

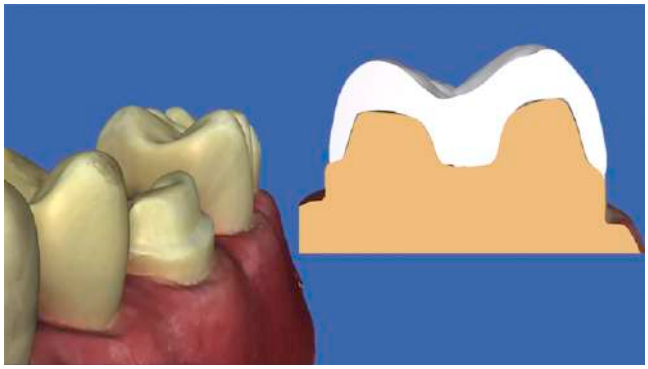


Figure 22.8 Ferrule margin design with opposing walls for mechanical retention more appropriate for cast metal restorations.

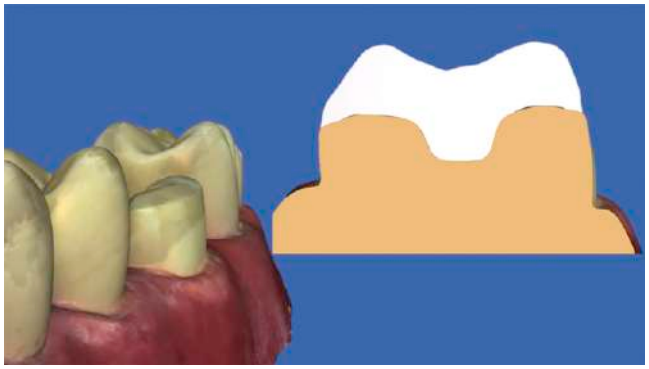


Figure 22.10 Modified butt-joint margin with a football-shaped diamond for improved esthetic blending of the onlay margin.





Figure 22.11 Chairside CAD/CAM workflow using the CEREC system. (A) Preoperative view of a mandibular first molar. (B) All-ceramic crown preparation for a mandibular first molar.

(*Continued*)



Figure 22.11 Cont. (C) Case identification in the Administrative Phase.



Figure 22.11 Cont. (D) Intra-oral digital scan of the mandibular left quadrant.

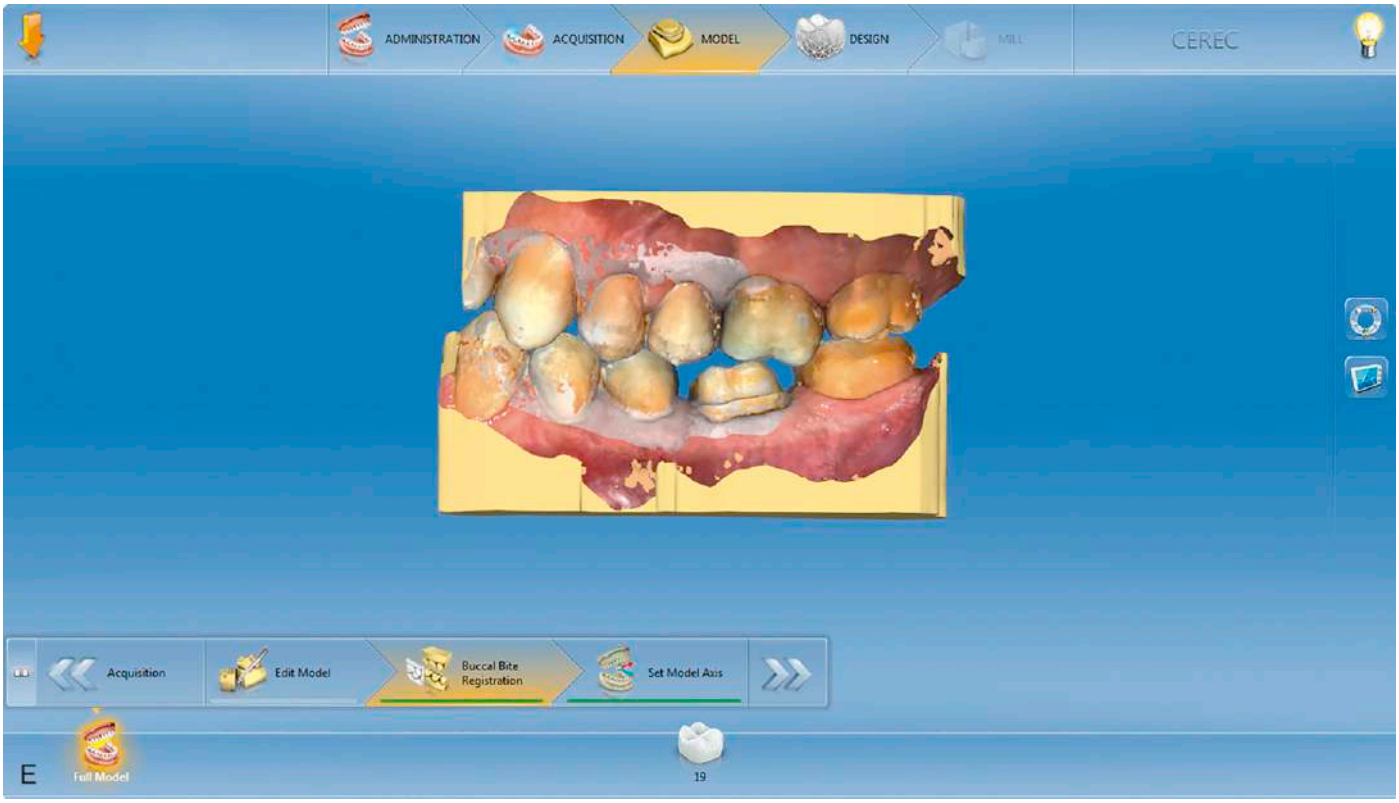


Figure 22.11 Cont. (E) Virtual mounting of the opposing digital models.



Figure 22.11 Cont. (F) The Model Axis is set for the model.

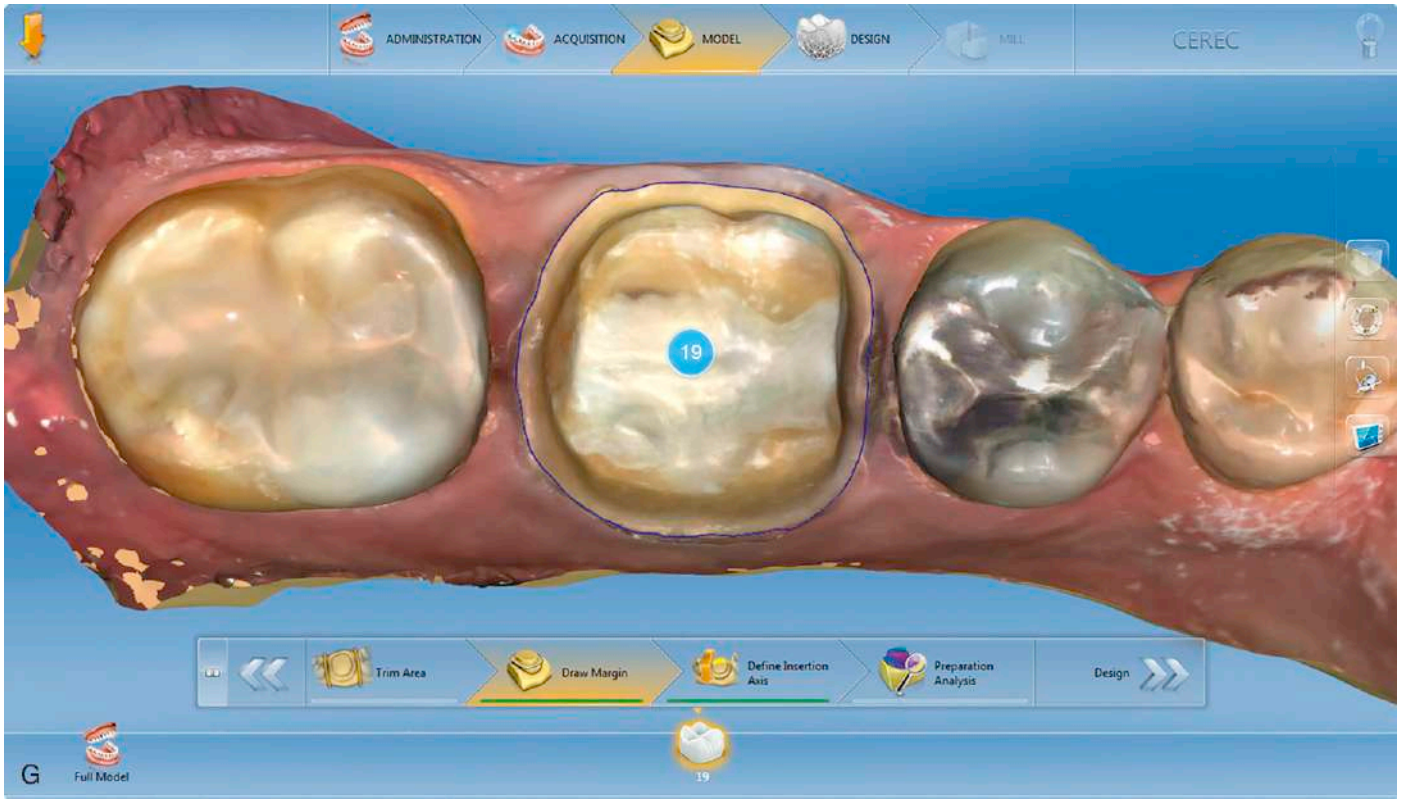


Figure 22.11 Cont. (G) The crown margin is identified.



Figure 22.11 Cont. (H) Biogeneric proposal of the crown.



Figure 22.11 Cont. (I) Visualization of the occlusal contacts for refinement.

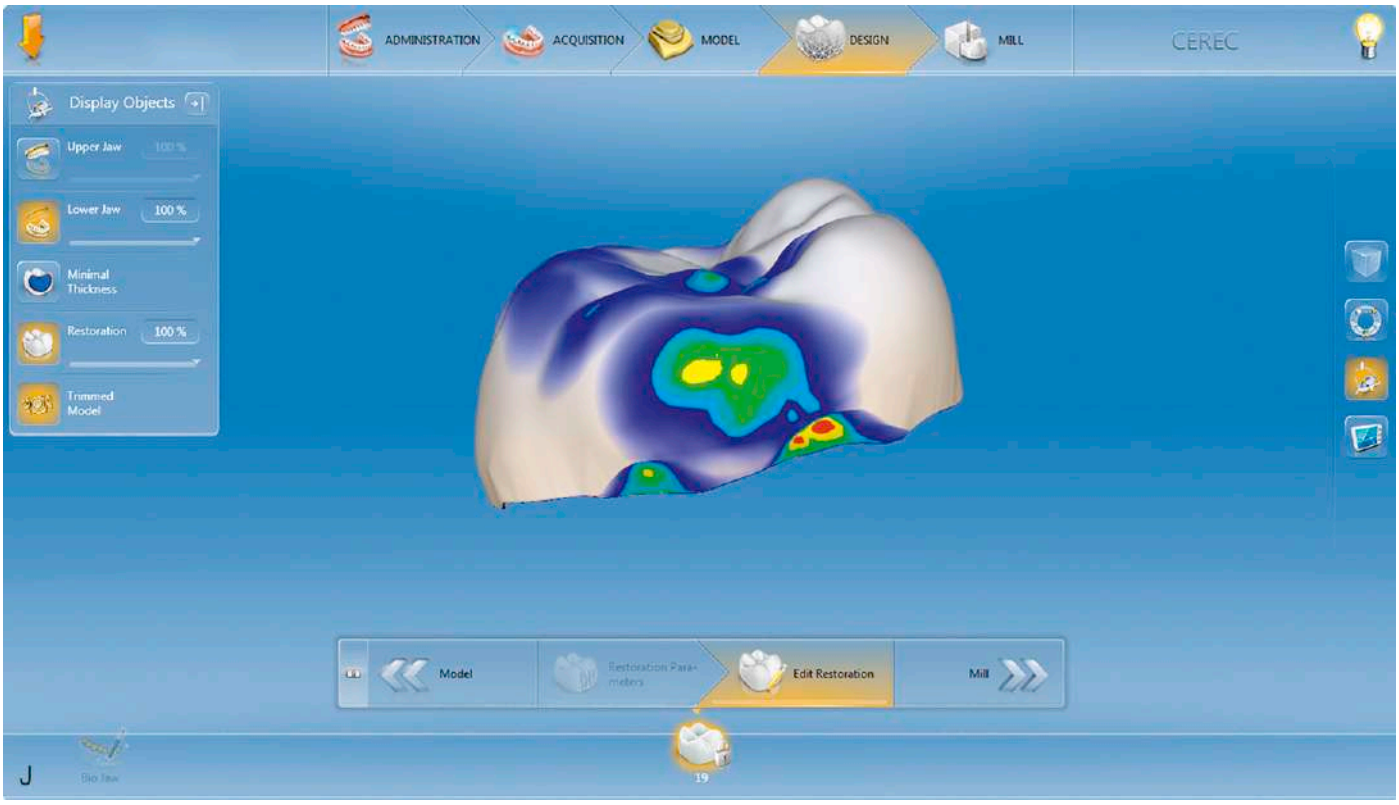


Figure 22.11 Cont. (J) Visualization of the proximal contact for refinement.

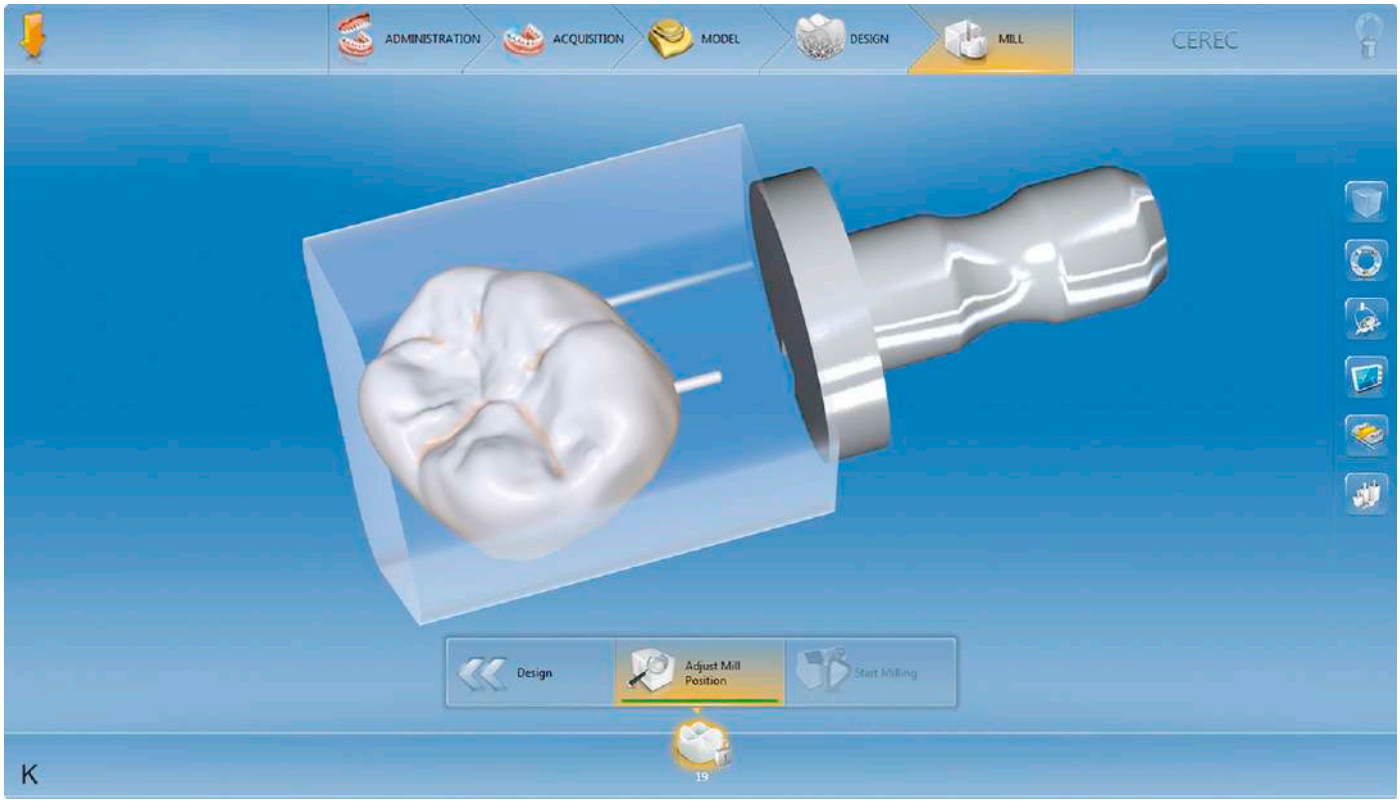


Figure 22.11 Cont. (K) Crown proposal in the proposed mill block (Celtra Duo).



Figure 22.11 Cont. (L) Delivered Celtra Duo ceramic crown for a mandibular first molar.

- The software utilizes data from adjacent and opposing teeth to generate a proposed (virtual) restoration, which may then be edited by a series of tools unique to each software program.
- The design software enables control of the emergence profile of the restoration, occlusal anatomy, planned occlusal contacts and size and intensity of proximal contacts.
- Once the design of the restoration is complete, it is transmitted to the milling unit so the fabrication process may create the volumetric shape of the restoration out of an industrially produced block of ceramic material.
- Although a variety of chairside fabrication processes may be imagined, subtractive grinding or milling processes of ceramic blocks are accomplished based on the volumetric design created with the software design program.
- Most restorations may be designed and milled within 10–15 min to allow for delivery of the restoration at the same appointment.
- Extended machining times may be required for more complex geometries of fixed partial dentures or implant restorations.
- Slower machining may be preferred for thin margins, such as for porcelain veneers, in order to prevent chipping during fabrication. Certain systems allow for fine machining modes, which slow the advancement of the block at each pass of the diamonds/burs (or allow for

a smoother second machining pass), so as to refine the surface contour.
- Final glaze firing and/or external surface polishing follows clinical confirmation of the anatomy and marginal adaptation of the resultant restoration.
- The final restoration is adhesively cemented with resin cement.

Chairside Restorative Materials

Dental material manufacturers produce *monolithic materials* for chairside CAD/CAM restorations generally referred to as *'blocks'*. The blocks are dense, homogeneous materials that have been industrially produced under ideal conditions so as to limit the presence of internal flaws (porosity or voids). Limited inclusion of flaws enables maximization of the physical properties of the material. The blocks are attached to mandrels that are specific to the various brands of chairside CAD/CAM milling units.

The first CAD/CAM block was created as a result of the collaboration between Dr Werner Mörmann and the Vita Corporation. The Vita Mark I blocks were originally created out of feldspathic porcelain and eventually evolved into the current generation of feldspathic blocks, Vita Mark II. Since then the CAD/CAM market has expanded to include multiple formulations of ceramic materials.

Chairside CAD/CAM materials may be divided among a number of categories based on material composition for ease in understanding their properties and clinical applications.[11,12] These categories include:

I. *Adhesive ceramics* (feldspathic, leucite-reinforced) that must be etched and adhesively bonded to the tooth structure.
II. *High-strength ceramics* (lithium disilicate, zirconia reinforced lithium silicate) with improved strength properties compared to the adhesive ceramic materials.
III. *Resilient ceramics* that do not require a porcelain furnace and must be adhesively bonded to the tooth.
IV. *Composite materials* that have a resin matrix.
V. *Full contour zirconia* that may be cemented to the tooth.
VI. *Provisional materials* used for temporary restorations (Table 22.2).

I. Adhesive Ceramics

The adhesive ceramic category includes both fine-grained feldspathic porcelain and leucite-reinforced porcelain.

Feldspathic Porcelain

- These materials contain a significant glass component resulting in increased translucency allowing them to have an improved *'chameleon'* effect. The chameleon effect may be described as the ability to reflect the colour of the surrounding tooth structure and thus blend into the existing tooth shade (Fig. 22.12).
- These materials also have a moderate flexural strength on the order of 100–175 MPa.
- The glass component of these materials allows them to be etched with hydrofluoric acid for micromechanical adhesive bonding. Adhesive bonding is critical to their long-term success because the glass matrix makes them brittle.
- These materials do not have enough inherent physical strength to be cemented with resin-modified glass ionomer or traditional glass ionomer cements.
- The adhesive resin cement not only provides retention for the restoration but, by virtue of its adhesive

Table 22.2 Restorative material options for chairside CAD/CAM systems

Material	Brand	CEREC	PlanFit	CS Solution
Adhesive ceramic				
Leucite-reinforced	IPD EmpressCAD (Ivoclar)	X	X	–
Feldspathic	Vita Mark II (Vita)	X	X	X
	Sirona Blocs (Dentsply Sirona)	X	–	–
High-strength ceramic				
Lithium disilicate	IPS emaxCAD (Ivoclar)	X	X	–
Zirconia-reinforced lithium-silicate	Celtra Duo (Dentsply Sirona)	X	–	–
Lithium-silicate	Obsidian (Glidewell)	–	X	–
Resilient ceramic (resin nano ceramic)	Lava Ultimate (3M)	X	X	X
Hybrid ceramic	Enamic (Vita)	X	X	X
	Cerasmart (GC America)	X	–	–
Composite resin	Paradigm MZ100 (3M)	X	–	–
	Brilliant Crios (Coltene)	X	–	–
Zirconia (full contour)	CEREC Zirconia (Dentsply Sirona)	X	–	–
Provisional acrylic materials	TelioCAD (Ivoclar)	X	X	–
	Vita CAD-Temp (Vita)	X	–	–

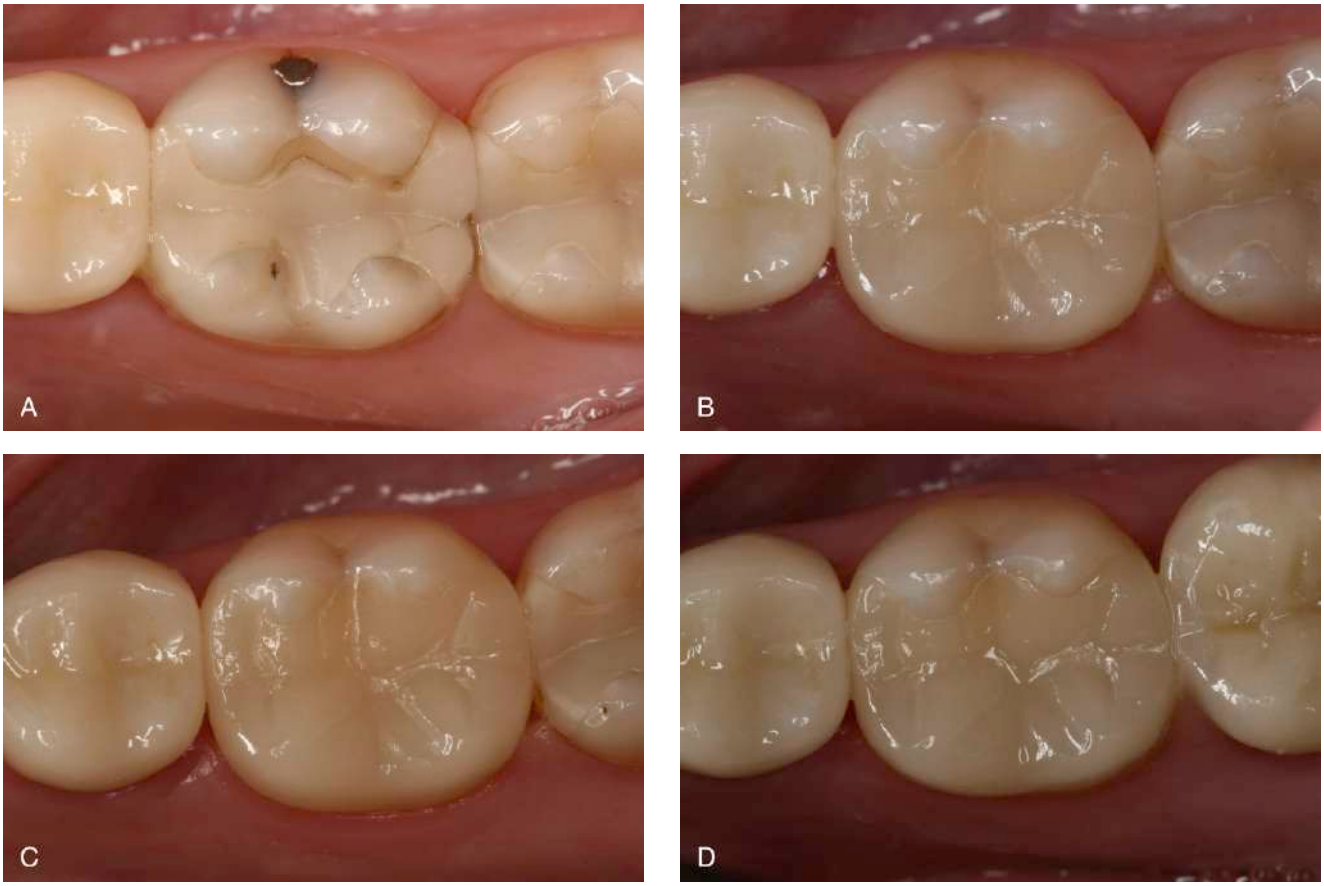


Figure 22.12 Vita Mark II (Vita) onlay for a mandibular first molar. (A) Preoperative view of a mandibular first molar. (B) Vita Mark II onlay delivered. (C) Vita mark II onlay at the 3-year recall. (D) Vita mark II onlay at the 5-year recall.

properties, contributes to the physical strength of the ceramic to resist fracture by limiting flexure under occlusal load.

- Vita Mark II (Vita) and Sirona Blocks (Dentsply Sirona) are both feldspathic porcelain blocks with a fine-grained particle size that averages 4 microns and a flexural strength of 100 MPa.[13,14]

Leucite-reinforced Glass Ceramic

- The leucite-reinforced glass ceramic was initially introduced as ProCAD (Ivoclar) and later evolved into IPS EmpressCAD (Ivoclar) with a similar crystal structure as IPS Empress 1 (Ivoclar). EmpressCAD has a finer particle size of 1–5 microns that is evenly dispersed 35%–45% by volume using a proprietary manufacturing process. It comes in two levels of translucency as well as multishaded blocks.[13]
- The manufacturer-reported flexural strength is 160 MPa, which is within the range of what is reported independently in the literature. EmpressCAD also has a light scattering behaviour that allows for a natural chameleon effect (Fig. 22.13).

II. High-strength Ceramics

There are several higher strength glass ceramic materials available for chairside CAD/CAM restorations.

IPS e.maxCAD (Ivoclar)

- This is composed of lithium disilicate and has a significantly greater flexural strength and fracture toughness than other adhesive glass ceramics.[15,16]
- The manufacturer provides the IPS e.maxCAD block in a partially crystallized state (160 MPa, 40% crystallized by volume) to allow for easier and more efficient grinding of the material. After the restoration has been fabricated, it must be subjected to a two-stage firing cycle in a ceramic furnace under vacuum to complete the crystallization process and achieve the maximum flexural strength potential of the material (500 MPa). The resulting glass ceramic restoration has a grain size of approximately 1.5 microns with a 70% crystal volume incorporated in a glass matrix. The crystallization firing also creates the optimum translucency for the material (Fig. 22.14).



Figure 22.13 IPS EmpressCAD (Ivoclar) restoration. (A) Onlay preparation for a mandibular first molar. (B) IPS EmpressCAD onlay delivered. (C) IPS EmpressCAD onlay at the 2-year recall.

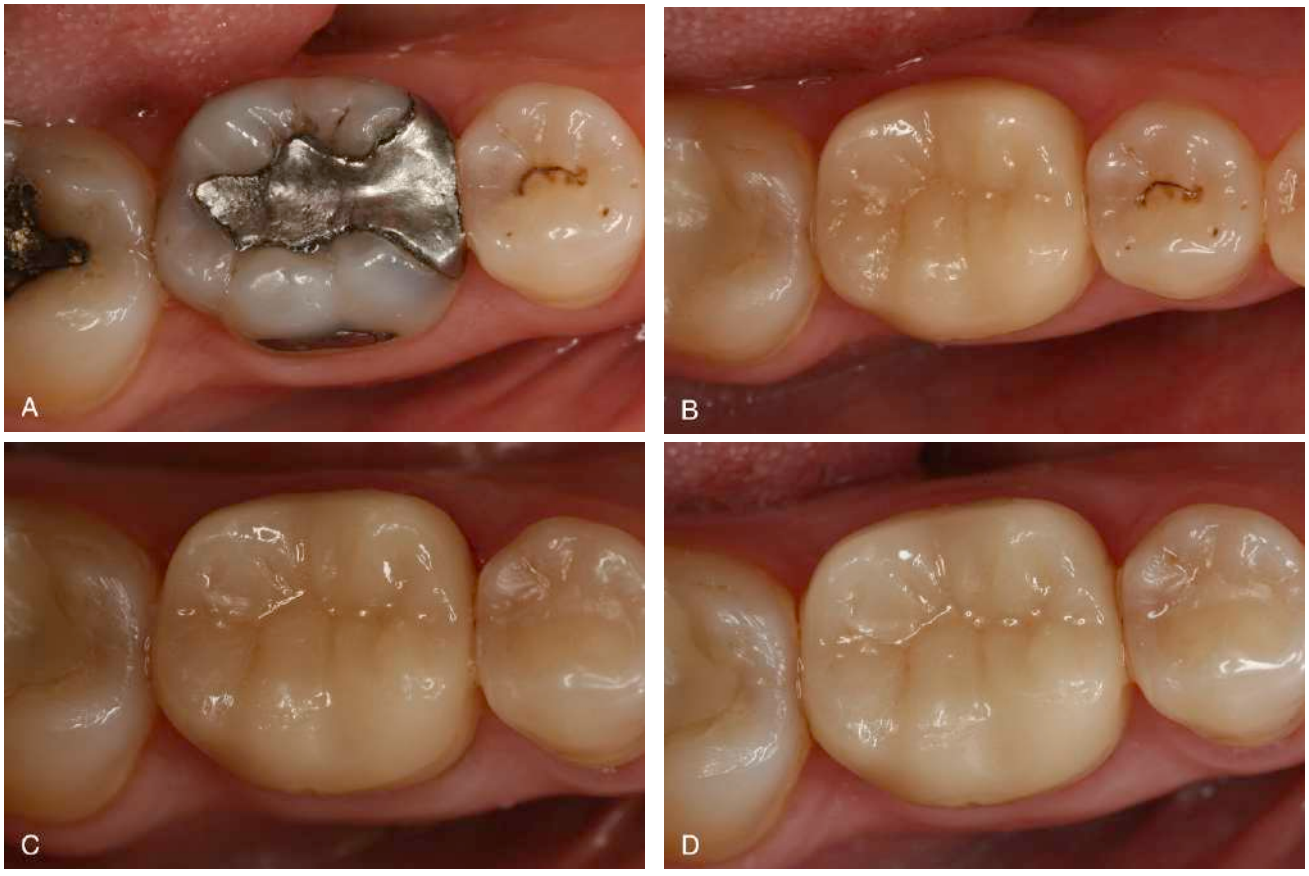


Figure 22.14 IPS emaxCAD (Ivoclar) crown for a mandibular first molar. (A) Preoperative view of a mandibular first molar. (B) IPS emaxCAD crown delivered. (C) IPS emaxCAD crown at 2-year recall. (D) IPS emaxCAD crown at 4-year recall.

Celtra Duo (Dentsply Sirona)

- This is a zirconia-reinforced lithium silicate (ZLS) material that is another example of ceramic with high strength. The ZLS microstructure has high content of ultrafine glass ceramic crystals (<1 μm) and 10% zirconia content. It is provided by the manufacturer in a fully crystallized state that may be either hand polished or glaze fired in a ceramic furnace prior to delivery. Hand polishing the restoration results in a material that has a flexural strength of 210 MPa, while glazing it in a porcelain oven results in a restoration with a flexural strength of 370 MPa. Celtra Duo is available in shades A1, A2, A3, A3.5 and B2 in both high and low translucencies (Fig. 22.15).

NICE (Straumann)

This is another fully crystallized ZLS block that has recently been introduced for chairside CAD/CAM restorations.

III. Resilient Ceramics

- The Resilient Ceramic category of chairside CAD/CAM blocks (Lava Ultimate [3M], Enamic [Vita] and Cerasmart [GC America]) includes materials that have a *resin matrix* (instead of a glass matrix) that, according to manufacturers, allows *greater force absorption* capability without fracture.
- The added feature of these materials is that *no additional firing* is required allowing for a very efficient delivery process following fabrication of the restoration.
- All three materials (Lava Ultimate, Cerasmart and Enamic) exhibit *flexural strength properties similar to human dentin*, and are indicated for singe-unit restorations.
- In contrast, the low stiffness properties of these materials may be considered disadvantageous from a bonding perspective, as marginal seal debonding due to material flexure might occur.[17] Therefore, treating the intaglio surface of the restoration with sandblasting technique as well as selecting a *total etch approach combined with adhesive cementation* is considered to be imperative.
- As resilient ceramics are less dense they mill faster, with a smaller incidence of margin chipping during milling compared to glass-containing materials.[17]

Lava Ultimate (3M)

This is a nanoceramic material that contains 20 nm size silica particles, 4–11 nm size zirconia particles and



Figure 22.15 Glazed Celtra Duo (Dentsply Sirona) crown for a maxillary first and second premolar. (A) All-ceramic crown preparations for a maxillary first and second premolar. (B) Facial view of all-ceramic crown preparations from the facial. (C) Delivered glazed Celtra Duo crowns. (D) Facial view of delivered Celtra Duo glazed crowns.

agglomerated nanosize particles of silica and zirconia, all embedded in a highly cross-linked polymer matrix with an approximately 80% ceramic load. Lava Ultimate has a reported flexural strength of 170 MPa and it is indicated for inlays and onlays but not for crowns (Fig. 22.16).[17,18]

Cerasmart (GC America)

This is described by the manufacturer as a flexible nanoceramic with a resin matrix containing homogeneously distributed nanoceramic filler particles. The material is a high-density composite resin with 71% silica and barium glass nanoparticles filler by weight.[17] The reported flexural strength of Cerasmart is 230 MPa and it is indicated for inlays, onlays and crowns.[17]

Enamic (Vita)

This is described by the manufacturer as a resin-based (14 wt%) hybrid ceramic comprised of a dual interpenetrating structure of a leucite-based and zirconia-reinforced ceramic network (86 wt%). The mechanical properties of the material are in between glass ceramics and highly filled composites.[19] The ceramic network provides wear resistance; however, it makes the material more brittle and susceptible to fracture. The polymer network is capable of improving the fracture resistance of the material due to its capability of undergoing plastic deformation.[20] The manufacturer indicates the material for inlays, onlays and crowns and reports that flexural strength of Enamic is 150 MPa, which is significantly lower than Lava Ultimate or Cerasmart. This is in line with the results of an independent study (approximately 135 ± 25 MPa).[17]

IV. Composite Materials

Composite resin block materials are gaining in interest as the efficiency of chairside designing and milling a CAD/CAM restoration has improved. The chairside CAD/CAM workflow allows for easier control of proximal contours, contacts and occlusal relationships for larger, more complex multi-surface composite restorations compared to incremental hand placement of direct composite materials.

Paradigm MZ100 (3M)

This is a composite block based on Z100 composite chemistry and relies on a proprietary processing technique to maximize the degree of cross-linking in the Bis-GMA polymer-based composite material. It has zirconia–silica filler, which is radiopaque, and is 85% filled by weight with an average particle size of 0.6 micron.[21] The manufacturer reported flexural strength of Paradigm MZ100 is 150 MPa. An independent study reported slightly higher value 157 ± 30 MPa, which was not significantly different than the flexural strength reported for EmpressCAD.[17]

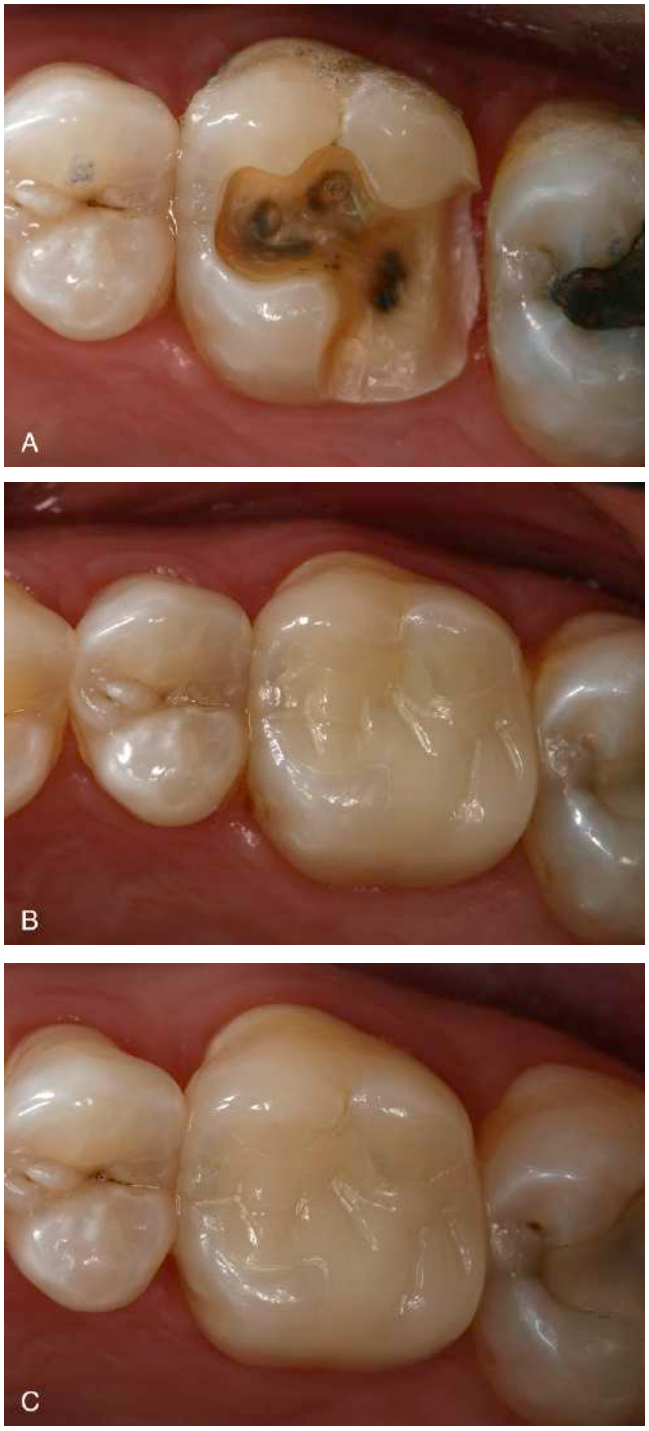


Figure 22.16 Lava Ultimate (3M) onlay for a maxillary first molar. (A) Onlay preparation for a maxillary first molar. (B) Lava ultimate onlay delivered. (C) Lava Ultimate onlay at the 3-year recall.

Brilliant Crios (Coltene)

This is a recently introduced reinforced composite block-containing amorphous silica and glass ceramic particles in a cross-linked methacrylate matrix.

V. Full Contour Zirconia

The most recent material innovation for chairside CAD/CAM restorations was the introduction, in 2016, of CEREC Zirconia (Dentsply Sirona).

CEREC Zirconia (Dentsply Sirona)

- This material is a precoloured, full-contour zirconia that may be processed efficiently for a single-appointment delivery. The fast production time is a result of the simultaneous introduction of the SpeedFire (Dentsply Sirona) furnace that is able to sinter the zirconia restorations in under 20 min.
- Zirconia has a flexural strength and fracture toughness that is generally at least three times as great as glass ceramic materials.[22] CEREC zirconia has a manufacturer-reported flexural strength of over 900 MPa; therefore it is *suitable for crowns* as well as *short-span fixed partial dentures* (*FPDs*) that may be cemented with traditional cements.
- A unique property of zirconia is that it *shrinks volumetrically approximately 22%–24% when sintered*. The CEREC software program *automatically compensates* for the degree of shrinkage in the milled restoration. When the restoration design is ready for milling, the bar code on the zirconia block is input into the software program. The bar code contains the precise amount of expected shrinkage of the particular zirconia block as determined by the manufacturer. The software virtually expands the designed restoration by this degree and an enlarged restoration is milled. The CEREC software automatically programs the SpeedFire furnace to sinter the milled zirconia restoration and firing shrinkage results in a final restoration that accurately fits the preparation.
- The sintering process takes 10–15 min for crowns and more than 25 min for FPDs. Subsequent glaze firing and surface colour characterization is optional. Glazing of the zirconia restoration may be accomplished in the SpeedFire furnace or in a conventional porcelain furnace.

VI. Provisional Materials

CAD/CAM blocks for provisional restorations are also available for chairside fabrication of long-term provisional crowns and FPDs. The chairside CAD/CAM process avoids the air-inhibited layer present on conventional self-cure or VLC acrylics as well as polymerization shrinkage.

Vita CAD-Temp Blocks (Vita)

These are made from a highly cross-linked, microfilled polymer and are available in extended block sizes, including 40 and 55 mm lengths, to accommodate multiple unit FPDs.

Telio CAD (Ivoclar)

This is a millable cross-linked polymethyl methacrylate block that is part of the Telio System (a self-curing composite, desensitizer and cement). The Telio CAD material is also available in 40 and 55 mm size blocks.

Accuracy of Digital Impressions

It is axiomatic that the accuracy of the final restoration depends on the accuracy of the recorded dimensions of the tooth preparation. This is true for both conventional and digital impressions. The accuracy of the margin fit and internal adaptation of any restoration is limited by the geometry of the tooth preparation and the limitations of the recording medium. The tooth preparation must be well isolated from moisture contamination and adjacent soft tissues if an accurate digital impression is to be made.

Current technology will not allow the digital capture of the tooth preparation through saliva, blood or soft tissue. A simple way to think about digital impressions is that the camera can only accurately record what is clearly visible to the eye of the operator. Careful control of the scanning environment ensures an accurate surface reproduction with the digital impression.

Conventional impression materials require lateral and cervical retraction of the soft tissues for an accurate impression. Lateral space must be adequate to ensure sufficient impression material to resist tearing at the margin. Cervical space, generally 1 mm cervical to the margin, is required for accurate assessment of the recorded margin in the impression. Digital impressions have the advantage that soft tissues only need be retracted laterally in order to visualize the margin.

Digital impressions provide excellent immediate feedback relative to the recorded tooth preparation. Digital magnification of the image, in many cases up to 20 times lifesize, facilitates critical evaluation of the tooth preparation while the patient is still in the chair. Preparation corrections may be accomplished immediately. Additionally, inadequately captured areas may be immediately reimaged without the need to redo the entire impression, as is the case with conventional impression materials.

Research Relative to CAD/CAM Systems and Materials

Dentists have ongoing concern that restoration adaptation to tooth preparations allows for optimal clinical performance. The CEREC system was the initial chairside CAD/CAM system introduced to the dental marketplace and has over 30 years of both laboratory and clinical research documenting the relative accuracy of the fit of CEREC chairside CAD/CAM restorations.

Digital camera technology has evolved from a single image capture process (RedCam) to the LED-based BlueCam to the present video version (OmniCam). The fit of CEREC restorations have consistently been considered accurate irrespective to the version of the image capturing device used.

Marginal adaptation (or margin fit) plays a critical role in long-term clinical performance of CAD/CAM restorations; when the adaptation of the restoration to the preparation is poor it leads to greater discrepancies at the margin. Restoration adaptation has been thoroughly investigated in the literature. Although some internal space is necessary to accommodate the resin cement, *most authors are in agreement that margin openings under 100 µm would be desirable*.

- A systematic review evaluated the fit and margin adaptation of CAD/CAM restorations.[23]. The marginal gap of crowns, made from a variety of primarily ceramic materials, ranged from 10 to 110 µm, often with gaps less than 80 µm.
- Studies [24–28] assessing the marginal fit of restorations fabricated using digital impressions have found that difference in fit amongst systems was below the accepted clinical threshold of 100 µm and therefore it may not be clinically relevant.
- A study measured the ability of six intra-oral scanners to accurately capture a single molar abutment tooth in vitro.[29] The authors concluded that, even though there were differences in levels of accuracy, all scanners investigated produced clinically acceptable accuracy.
- A more recent systematic review also evaluated the marginal fit of single-unit all-ceramic crowns fabricated after conventional and digital impressions. The search included studies published from January 1989 through December 2014. The result of the meta-analysis indicated no statistically significant difference in marginal gap of single-unit ceramic restorations fabricated after conventional or digital impressions. The authors suggested that the *digital workflow exceeds clinically acceptable standards and performs equal to conventional impressions*.[30]
- The PlanFit and CS Solutions systems have been more recently introduced for chairside CAD/CAM restorations and have very limited published research on margin fit and internal adaptation. Additional independent studies are warranted for a more precise assessment of margin fit and internal adaptation of PlanFit and CS Solutions restorations.

Clinical Longevity of CAD/CAM Restorations

The clinical performance of chairside CAD/CAM restorations has been studied so as to assess relative longevity in the oral environment. Long-term randomized clinical trials are considered the most robust study design for the purpose of proper assessment of clinical longevity. However, there is considerable amount of variation in the dental literature when searching for independent randomized studies on clinical longevity of chairside CAD/CAM restorations. The vast majority of the publications report on the long-term clinical performance of CEREC restorations primarily because the CEREC System has been available since the early 1990s, whereas other systems have been more recently introduced. Systematic reviews represent invaluable tools to synthetize the results of these multiple and varied clinical studies; however, the conclusions of systematic reviews are only as valuable as the quality of the studies they seek to combine.

- In a systematic review,[31] the estimated survival rate for CEREC single-tooth restorations was 91.6% (95% CI: 1.22%–2.52%). Restorations fabricated with feldspathic porcelain had the highest 5-year survival rate, which contrasted with the lowest 5-year survival rate for glass ceramic. At 5 years, ceramic onlays performed equally as successful as crowns; however, the authors identified a significant lack of randomized studies as well as a shortage of scientific evidence longer than 3 years.
- Among the CEREC literature, several of the studies published data on CAD/CAM generated feldspathic porcelain restorations. A retrospective study on the clinical performance of 2328 inlays and onlays for 794 patients in a private practice setting reported a total of 35 failures over 9 years. The Kaplan–Meier survival probability reported was *97.4% at 5 years and 95.5% at 9 years for CEREC inlays and onlays*. According to the authors, failure rate was not significantly influenced by restoration size, tooth vitality, previous presence of deep caries, type of tooth treated or whether the restoration was located in the maxilla or mandible. The most common type of failure reported was tooth extraction.[32–34]
- The clinical survival of CAD/CAM restorations is influenced by the adhesive cementation, which in turn is maximized with the *presence of enamel margins*. This was well demonstrated by a longitudinal study on clinical performance of nonretentive feldspathic CAD/CAM overlays (occlusal veneers). A single operator placed 310 CEREC overlays paying particular attention to keeping preparation margins within enamel whenever possible. After 8 years of follow-up, 286 paired onlays were available for evaluation and the calculated survival rate was 99.3%. The only two fractures observed in this study were on maxillary premolars of one patient with occlusal parafunction.[35] The clinical success of these restorations was likely due to the predictable bonding that is often achieved whenever the all-ceramic preparation is within enamel. In addition, an experienced operator may have

influenced such remarkable performance, as more inexperienced operators did not achieve similar results.

There is considerable published clinical research on chairside CAD/CAM technology that has revealed it to be a valuable tool for the delivery of high-quality restorations delivered during a single appointment. The ability to adhesively bond ceramic restorations at the time of restoration fabrication minimizes the occurrence of postoperative sensitivity while allowing for more conservative tooth preparation as no preparation retention is required to retain a temporary restoration. Research studies have assessed an extensive variety of resin-based, porcelain, glass ceramic and high-strength ceramic materials that may be used to provide restorative solutions for inlays, onlays, veneers and full crowns.

Summary

The clinical application of CAD/CAM technology has developed through a continuous evolution of the hardware and software. A variety of restorative materials are available to offer alternative solutions for a variety of clinical problems. Laboratory and clinical research has documented the accuracy, longevity and effectiveness of CAD/CAM fabricated restorations. The efficient treatment workflow provides an opportunity for patient-centred treatment in one appointment. The evidence-based reliability and validity, as well as the perceived value of same-day dentistry, have helped secure a permanent spot for CAD/CAM dentistry in operative dentistry practice.

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